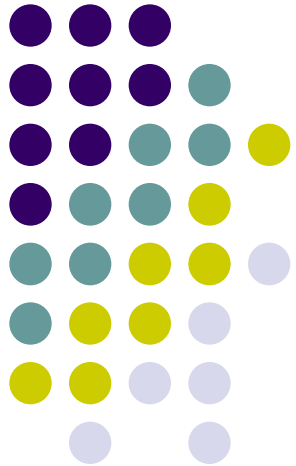
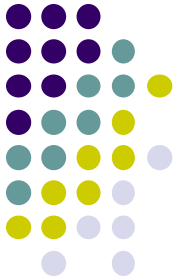
Session 9

Quantitative Data Collection





Session Outline

- Part 1
  - Key features of quantitative research
  - Sampling Strategies
  - Operational definitions
  - Types of quantitative research design
  - Types of quantitative data
  - Secondary data sources
  - Levels of measurement in qualitative research

- Part 2
  - Quantitative data collection methods
  - Pilot testing
  - Data collection planning and logistics
  - Ethical considerations
  - Common pitfalls

Learning Objectives

By the end of this session, you should be able to:

- Define quantitative research and its key characteristics
- Understand sampling strategies
- Identify different types of quantitative data
- Distinguish between the different sampling techniques and understand when to use each
- Understand and apply core data collection methods
- Design and evaluate a basic data collection tool
- Recognise common pitfalls and ethical considerations in quantitative research

Key Features of Quantitative Research

Numerical Data

Collects data in numerical form that can be measured and analysed statistically.

Objectivity

Aims to minimise researcher bias through standardised instruments and procedures.

Large Samples

Uses sufficiently large samples to support statistical inference and generalisation.

Replicability

Studies can be repeated under the same conditions to verify findings.

Hypothesis Testing

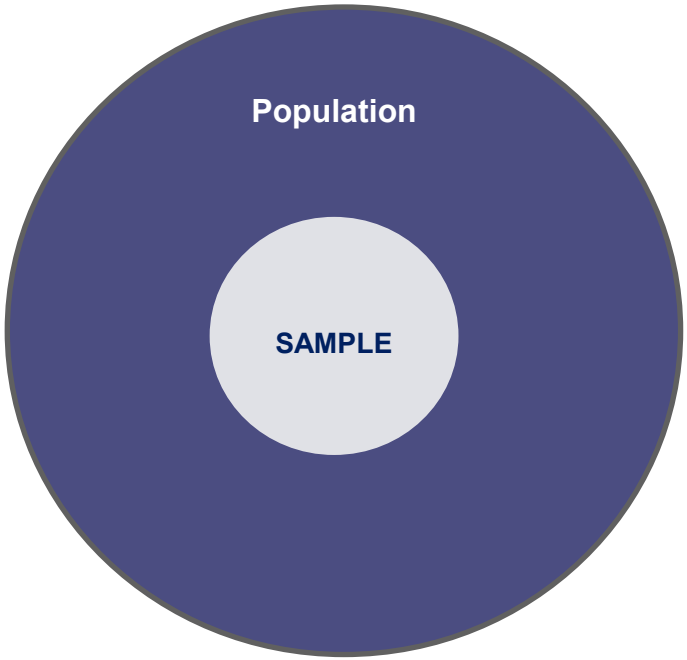
Tests pre-specified hypotheses using statistical methods to draw conclusions.

Structured Design

Follows a fixed research design with clear variables, instruments, and procedures.

Sampling

- Sample: Rather than studying an entire population, researchers analyse a carefully selected subset — a sample — to draw inferences about the wider group.
- Saves time and resources whilst providing meaningful, generalisable insights.
- Representativeness: How well the sample reflects the key characteristics of the target population



Sample Size

Sample Size

Too small: May fail to detect real effects (underpowered study), leading to inconclusive results.

Too large: Wastes resources and may raise ethical concerns — participants exposed to risk without sufficient justification.

Just right: Determined by a formal power calculation before the study begins, balancing feasibility and rigour.

Sampling Bias

Sampling Bias

- Occurs when the sample does not accurately represent the target population
- Leads to skewed or unreliable results
- Often caused by non-random selection methods
- Example: Surveying only gym-goers to study national physical activity overrepresents active people
- Example: An electoral roll may exclude younger adults, recent migrants, or the homeless — groups less likely to be registered voters

Probability Sampling Strategies

Each individual has a known, non-zero chance of selection

Simple Random

Every individual is **randomly selected** from the full population list

Example

Randomly select 100 patients from a hospital register

Ideal: least bias

Systematic

Select **every kth individual after a random start point**

Example

Select every 10th name from an alphabetically sorted list

Easy to implement

Stratified

Divide population into **strata** (subgroups) and **randomly sample from each stratum**

Example

Sample doctors and nurses separately to ensure both groups are represented

Ensures subgroup representation

Cluster

Divide population into **clusters**; **randomly select clusters**, then **sample within the selected clusters**

Example

Randomly select 5 hostels, then survey students within each

Cost-effective for dispersed populations

Which Probability Sampling Strategy to Use?

Characteristic	Simple Random	Systematic	Stratified	Cluster
Use Case	Complete list, equal chance	Ordered list, simplify selection	Ensure subgroup representation	Large, spread out population
Example	Select 100 patients from registry	Select every 10th patient	Sample doctors and nurses separately	Select hospitals, then sample patients

Non-Probability Sampling Strategies

Participants chosen non-randomly — some may have zero chance of selection

Convenience

Participants are selected because they are **easily accessible**

Example

Surveying students in the library during lunch break

Best used when:

Early-stage or exploratory research

Purposive

Participants are deliberately chosen because they **meet specific research criteria**

Example

Selecting experienced GPs to study clinical decision-making

Best used when:

specialist knowledge or characteristics are needed

Quota

Participants are chosen to **fill pre-specified quotas representing key subgroups**

Example

Ensuring 50 male and 50 female respondents in a study on health behaviours

Best used when:

Proportional representation of subgroups is important

Snowball

Existing participants recruit further participants from their networks

Example

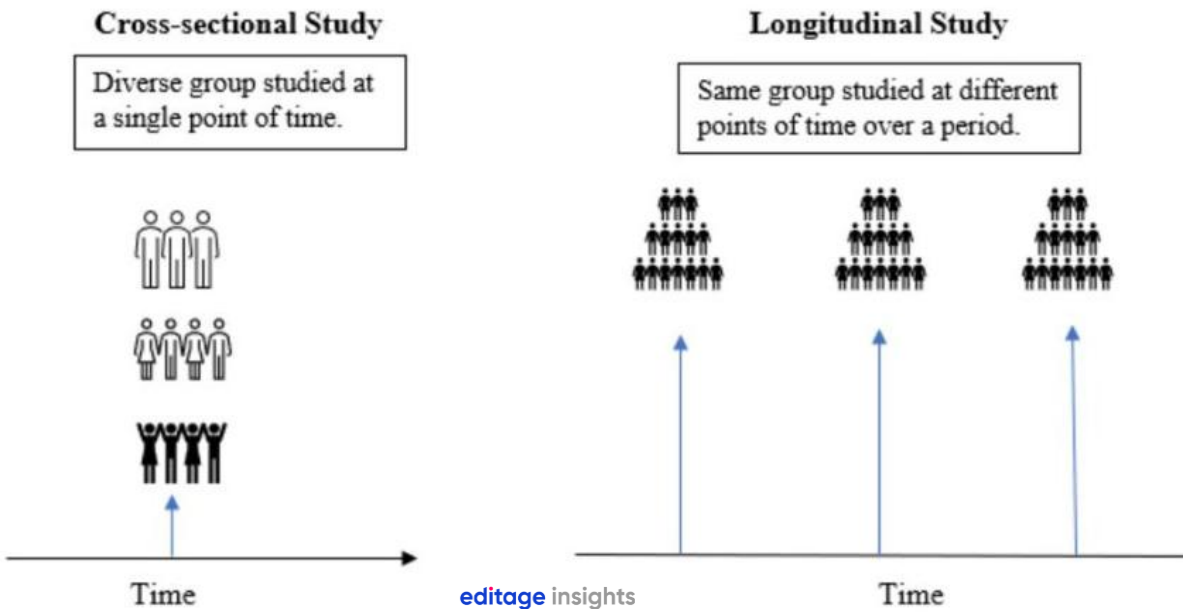
Recruiting homeless individuals for a study on housing needs via peer referral.

Best used when:

Hard-to-reach or hidden populations are targeted

⚠ Results are non-generalisable — findings may not apply to the wider population. Representativeness is traded for accessibility or practicality.

Types of Quantitative Data



Cross-sectional Data: Data collected at one point in time

Longitudinal Data: Data collected over multiple time points

Types of Quantitative Data

Cross-sectional vs. Longitudinal

Cross-sectional Data

- Data collected at one single point in time
- Often used for prevalence or descriptive studies
- Quick to conduct and less costly
- Cannot establish causality — no follow-up
- Example: Surveying 500 adults about diet habits in October 2024

VS

Longitudinal Data

- Data collected over multiple time points
- Tracks changes and developments over time
- Useful for causal inference and trend analysis
- Time-consuming and subject to higher participant attrition
- Example: Tracking children's reading ability every 6 months over 3 years

Primary vs Secondary Data

Primary Data

- **First-hand data collected by the researcher**
- Tailored precisely to the research question
- Higher degree of control over collection method
- More reliable and purpose-fit
- Time-intensive and costly
- **Example:** Conducting your own survey on student anxiety

VS

Secondary Data

- **Second-hand data collected by others for different purposes**
- No control over how/why data was collected
- Comparatively less reliable — requires critical appraisal
- Time- and cost-effective
- May need cleaning or harmonisation before use
- **Example:** Using NHS hospital admission records for research

Secondary Data Sources

- Data originally collected for other purposes
- Incredibly useful for large-scale or resource-limited research projects

  National surveys

  - Conducted by governments/statistical agencies (e.g., census, health surveys)
  - Large sample sizes, nationally representative

  Administrative records

  - Data collected through service delivery (e.g., hospital admissions, school attendance)
  - Rich, longitudinal, and often real-time

  Open datasets

  - Publicly accessible datasets (e.g., WHO, World Bank, UN, OECD)
  - Useful for cross-country or trend analysis

Challenges of Secondary Data Sources

Whilst valuable, secondary data presents several quality and comparability challenges that must be address:

- Missing data e.g., incomplete income fields in health surveys
- Outdated information e.g., census data from 2013 used in 2020
- Inconsistent data entry e.g., “Male” vs “M” vs “1”
- Lack of documentation: unknown sampling or collection methods
- Changing definitions e.g., shift in “employment” criteria over time
- Non-standardised units e.g., monthly income vs yearly income vs brackets
- Cultural differences terms like “household head” interpreted differently

Operational Definitions

Operational definitions translate abstract concepts into precise, measurable observations — ensuring clarity, consistency, and replicability across the research process.

Example

Research topic on "stress"

Conceptual definition: Stress is a state of mental or emotional strain resulting from demanding circumstances.

Operational definition: **Stress** is measured using the **Perceived Stress Scale (PSS)**, with **scores ranging from 0 (no stress) to 40 (extreme stress).**

Clarity	Replicability	Objectivity	Precision
Reduces misunderstanding or misinterpretation of questions and variables.	The study can be repeated by others using the same measures.	Removes ambiguity and minimises researcher bias.	Supports consistent data collection and meaningful analysis.

Levels of Measurement in Quantitative Research

- How variables are measured influences data analysis
- The level of measurement determines the choice of appropriate statistical techniques
- Considering the four fundamental levels of measurement is essential

Nominal (N)	Ordinal (O)	Interval (I)	Ratio (R)
Categories with no meaningful order	**Ordered categories, but gaps between ranks are NOT equal**	**Equal intervals between values, but NO true zero**	**Equal intervals AND a true zero (zero means 'none')**
Examples: • Blood type (A, B, AB, O) • Nationality	**Examples:** • Satisfaction ratings (Low/Medium/High) • Exam grades (A, B, C, D, E)	**Examples:** • Temperature in °C or °F • IQ scores	**Examples:** • Height, weight, age • Income in $ • Distance in km
Can only count; cannot rank or calculate averages	*Can rank, but arithmetic is not meaningful*	*Zero does not mean 'absence' — 0°C is not 'no temperature'*	*All arithmetic operations are meaningful*

Levels of Measurement in Quantitative Research

- To properly analyse data, it is essential to understand how the data is measured.
- The four fundamental levels of measurement are:
  - Nominal: Categories without any order (**names or labels**)
  - Ordinal: Categories with a meaningful order, but no equal intervals (**order matters, but equal spacing cannot be assumed**)
  - Interval: Equal intervals, but no true zero (**0°C does not mean 'no temperature'**)
  - Ratio: Equal intervals and a true zero (**This is the most informative scale — all mathematical operations are valid**)

1. Annual income in NZD
A. Nominal
B. Ordinal
C. Interval
D. Ratio

2.Types of cuisine: Italian, Thai, Mexican
A. Nominal
B. Ordinal
C. Interval
D. Ratio

3.Student rankings: 1st, 2nd, 3rd
A. Nominal
B. Ordinal
C. Interval
D. Ratio

4.Temperature in Celsius
A. Nominal
B. Ordinal
C. Interval
D. Ratio

Types of Quantitative Research Design

1. Descriptive

What:

Describes characteristics, frequencies, or patterns in a population.

How:

No manipulation of variables — simply observes and measures what exists.

Example: Determining the percentage of UK university students who report experiencing academic stress.

Answers: What is happening?

2. Correlational

What:

Examines the relationship or association between two or more variables.

How:

No manipulation or control of variables — measures naturally occurring relationships.

Example: Examining the relationship between hours of sleep and academic performance in A-level students.

⚠ Correlation does NOT imply causation!

3. Experimental

What:

Establishes cause-and-effect relationships between variables.

How:

Involves random assignment to groups, manipulation of the independent variable, and control of extraneous factors.

Example: Testing a new cognitive behavioural therapy (CBT) programme vs. control group for anxiety.

Gold standard for causal claims — Answers: Why is it happening?

Types of Quantitative Research Design

Descriptive Research

Summarises characteristics of a population — averages, distributions, prevalence. Example: national census data.

Exploratory Research

Identifies patterns and generates ideas.

Estimation Studies

Estimates parameters (means, prevalence) with confidence intervals. Bayesian approaches fit here — oriented toward estimation.

Predictive Modeling

Regression, ML algorithms, and forecasting models. Focus is on prediction accuracy.

Hypothesis Testing

Null hypothesis significance testing (NHST). Common in medicine, psychology, and social sciences — but just one tool among many.

Correlational Research

Describes strength of relationships between variables. Reveals associations, not causation.



Quantitative Data Collection Methods

1. Surveys and Questionnaires

- Surveys and questionnaires gather structured data.
- Types of questions include:

  Multiple choice questions: Select one or more responses from options – Suitable for quick categorisation of data

  Likert scales: Measure attitudes or opinions (e.g., Strongly Agree – Strongly Disagree) – Suitable for gauging intensity of opinions; ideal for satisfaction or agreement

  Ranking: Priorities items by order of importance or preference – Suitable for understanding preferences (Caution: Too many items in ranking can frustrate respondents)

Which distribution mode should be used?



Online

Cost-effective, scalable, and flexible timing

Face-to-Face

High response rate and allows clarification

Telephone

Quicker responses but may have lower response rates

Surveys and Questionnaires

Reliability

Reliability is about **consistency**; does the instrument (e.g., survey or test) give the same results under consistent conditions?

- Test-retest reliability: If the same test is given to the same participants a week later, does it yield similar results?

- Internal consistency: Are the items within a scale, measuring the same concept?
  Example: In a 5-item questionnaire that measures **perceived stress**:
  1. I feel **overwhelmed** by responsibilities
  2. I feel **nervous or anxious**
  3. I have **trouble relaxing**
  4. I feel **under pressure**
  5. I feel **calm and in control**

  High internal consistency means people who strongly agree with Item 1 also tend to agree with Items 2, 3, and 4 and people who agree with Item 5 should disagree with the other items
  Low internal consistency may indicate that items may not be measuring the same concept possibly due to poor wording or unrelated content (bad design).

Surveys and Questionnaires

Validity

- Validity refers to whether an instrument **measures what it is intended to measure**
- Example: If a questionnaire is developed to measure **anxiety** but primarily includes items related to "fatigue and appetite!!", its validity may be questioned because core components of anxiety such as **worry, nervousness, and restlessness** are not adequately captured.
- For a measurement to be considered valid, it must first demonstrate reliability

2. Experiments and Trials

Experiments and trials are controlled investigations to determine cause-and-effect relationships

- Involve manipulation of independent variables to observe changes in dependent variables
- Often use control and experimental groups

Example: Does a new app improve medication adherence?

Researchers want to find out whether using a medication reminder app causes patients to take their medication more regularly.

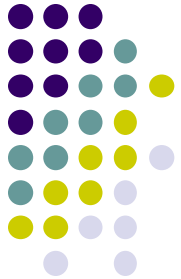
- **Independent variable:** Use of the medication reminder app
- **Dependent variable:** Medication adherence (e.g., percentage of prescribed doses taken)
- **Experimental group:** Patients who use the medication reminder app
- **Control group:** Patients who do not use the app and continue with usual care

After 8 weeks (for example), researchers compare medication adherence between the two groups.

If the experimental group has significantly higher adherence, this provides evidence that the reminder app may have caused the improvement.

3. Structured Observation

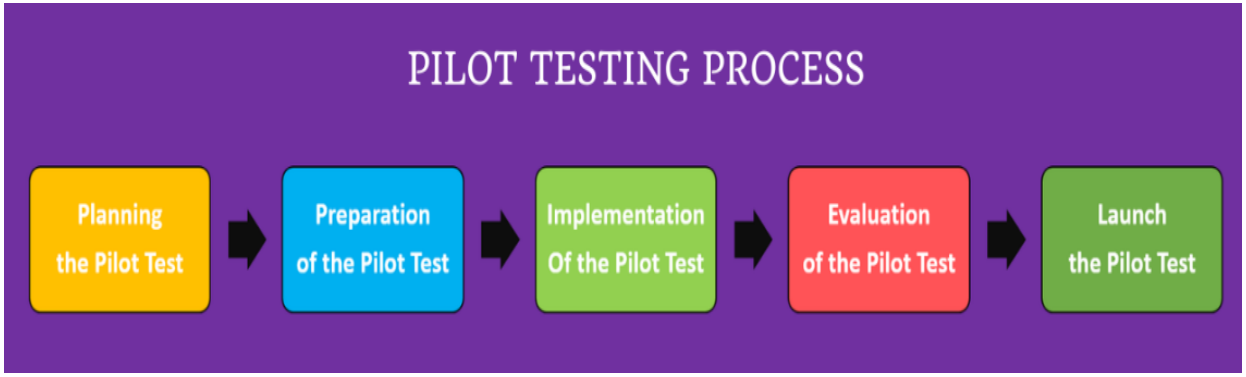
- A systematic method of watching and recording behaviours
- Used when behaviour must be quantified (e.g., workplace research, classroom studies)
- Unlike unstructured observation, observers follow a **predefined structure: what to observe, when, how**
- Use of Checklists and Coding Frameworks
  - Checklists: Predefined list of behaviours/events to look for, e.g., Did the student raise their hand? [✓/✗]
  - Coding Frameworks: Assign numerical or categorical codes to observed behaviour, e.g., "Interaction Type" → 1 = Verbal, 2 = Non-verbal, 3 = No interaction



Pilot Testing

Pilot Testing

- A **small-scale preliminary study** conducted **before** the main data collection to:
  - Test the reliability, clarity, and functionality of data collection tools (e.g., surveys, questionnaires)
  - Identify ambiguous or biased questions
  - Assess timing, flow, and respondent burden
  - Detect logistical or technical issues (e.g., online form glitches)



https://www.softwaretestingmaterial.com/pilot-testing/

Data Collection Planning and Logistics

1. Designing a Data Collection Plan

- Defining research objectives and variables
- Choosing suitable data collection methods (e.g., surveys, interviews)
- Setting a realistic timeline and resource plan
- Identifying data sources and sampling strategy

2. Piloting Data Collection Tools

- Testing instruments on a small group
- Identifying issues in question wording, timing, and functionality
- Revising tools before full-scale rollout

3. Managing Missing Data

- Planning for incomplete responses in advance
- Using reminders or mandatory fields (for online tools)
- Documenting reasons for missing data when possible

4. Data Storage and Security

- Storing data in secure, access-controlled platforms
- Encrypting sensitive files and anonymising personal identifiers
- Following institutional and legal data protection guidelines
- Backing up data regularly using secure methods

Ethical Considerations

- Informed Consent – Participants should voluntarily agree to participate after being clearly informed of the study's purpose, procedures, risks, and their rights

- Anonymity and Confidentiality – Protecting participants privacy
    - **Anonymity:** No identifying information is collected at all
    - **Confidentiality:** Data is collected but securely stored and shared only with authorised personnel

- Vulnerable Populations – Ensuring participation is truly voluntary and risks are minimised: Take extra care when involving minors, people with disabilities, marginalised groups, or others at risk of coercion (acting against their will)

- Ethical Approval – **Protects both the participants and the researcher**
    - Ethics approval must be obtained from a recognised ethics review board before data collection begins (E.g., EA1)

Common Pitfalls in Quantitative Data Collection

- Low response rates
  - ➤ Limits representativeness and statistical power
  - ➤ Mitigate with follow-up reminders, incentives, and clear communication
- Social desirability bias
  - ➤ Participants respond in a way they think is socially acceptable
  - ➤ Reduce with anonymous surveys and neutral wording
- Measurement error
  - ➤ Inaccurate or inconsistent responses due to poorly defined questions or tools
  - ➤ Use pilot testing and validated instruments to reduce error
- Poorly constructed instruments
  - ➤ Ambiguous wording, leading questions, or complex scales
  - ➤ Troubleshoot with expert review and pretesting